

LIPOPROTEIN RECOGNITION AND TRANSPORT

BY LOLCDE ABC TRANSPORTERS

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ABSTRACT

Antibiotic resistance is a major public health concern. Among the defence mechanisms of bacteria, the cell envelope acts as a first line of defence against antibiotics ¹. In Gram-negative bacteria, lipoproteins play an essential role in maintaining membrane integrity, adhesion, virulence and nutrient uptake. The transport of lipoproteins to the membrane is carried out by the specialised system ‘Lol’ (Localization of Lipoproteins). It consists of five proteins in *Escherichia coli*.

Lipoproteins are extracted from the inner membrane by an ABC transporter, LolCDE, then passed to the periplasmic chaperone LolA and finally inserted into the outer membrane by LolB ². The ABC transporter LolCDE represents a particularly attractive target for antibiotics. Recent structural advances have improved our understanding of its mechanism of action ³⁻⁵. However, many functional particularities exist in other Gram-negative bacteria, particularly in lipoprotein recognition, and still raise questions.

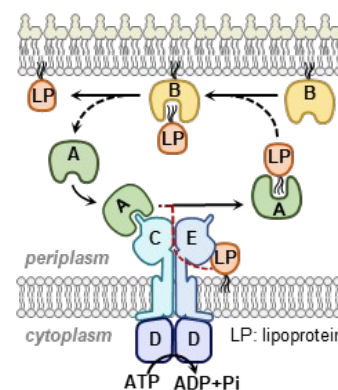


Fig. Lipoprotein sorting in *E. coli*.

The development of biochemical tools is essential for the functional and structural study of the Lol system. Various cloning, expression and purification strategies have been implemented, enabling the production of high-quality recombinant proteins. Initial characterisation shows that the purified LolCDE membrane complexes retain ATPase activity. In parallel, several protocols for reconstitution in proteoliposomes have been developed to reproduce lipoprotein transport *in vitro*.

These results represent a crucial step towards the complete reconstruction of the Lol system and the study of its action mechanism. The research carried out as part of this PhD thesis will draw on these tools to characterise the functional specificities of various LolCDE transporters and identify the factors involved in the recognition of lipoproteins. Altogether, these studies will assist the development of new antibacterial strategies targeting this essential pathway.

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